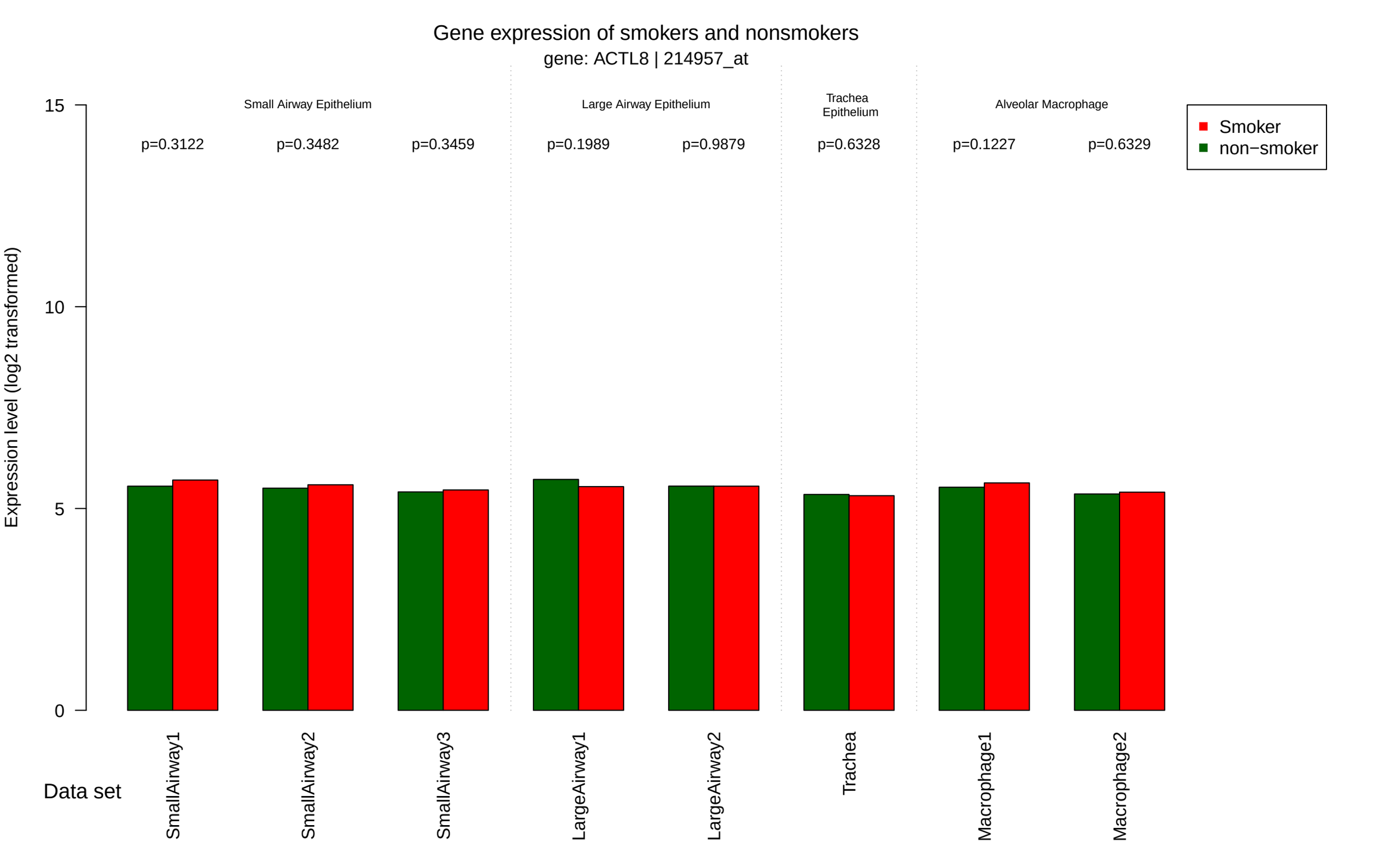
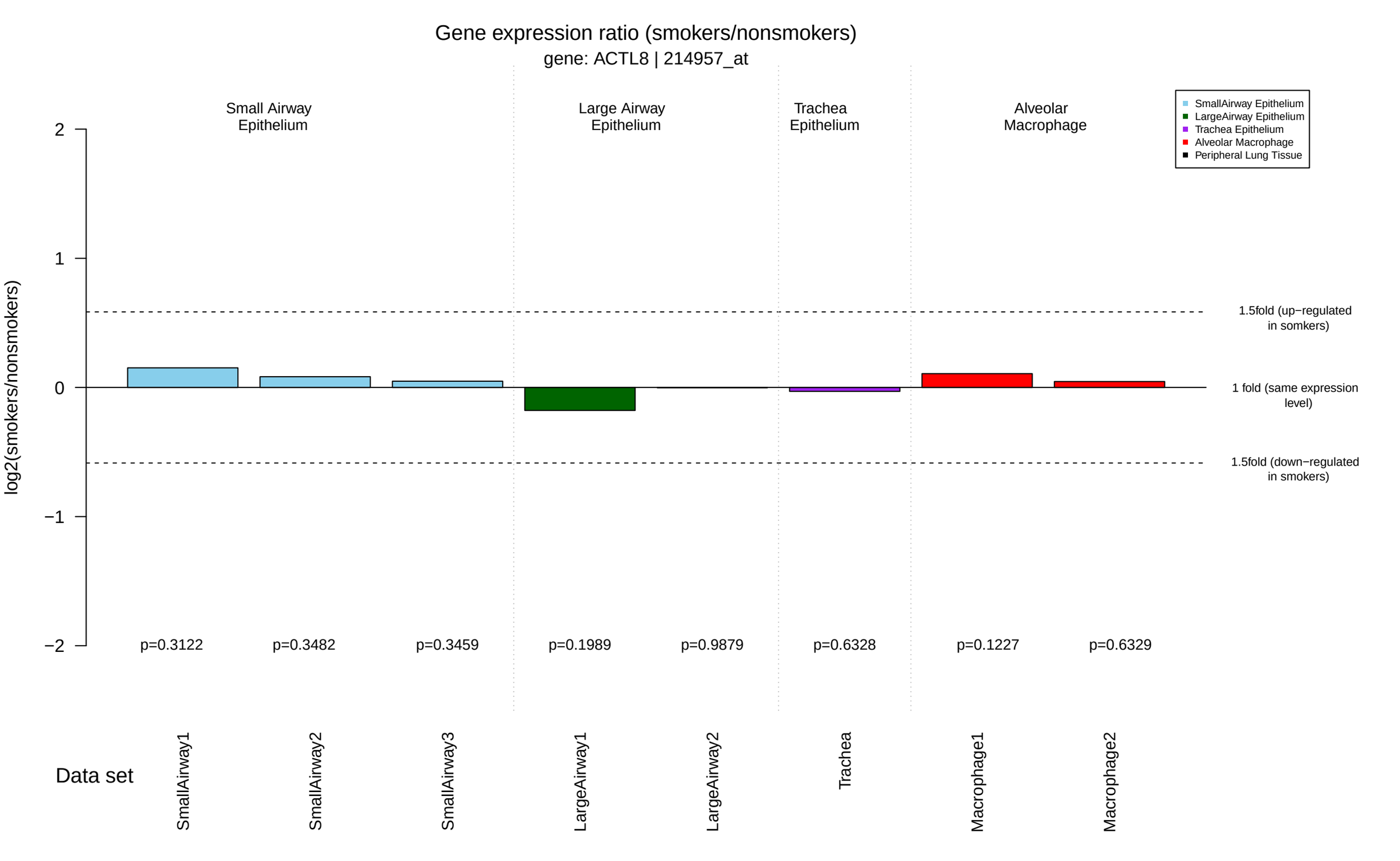
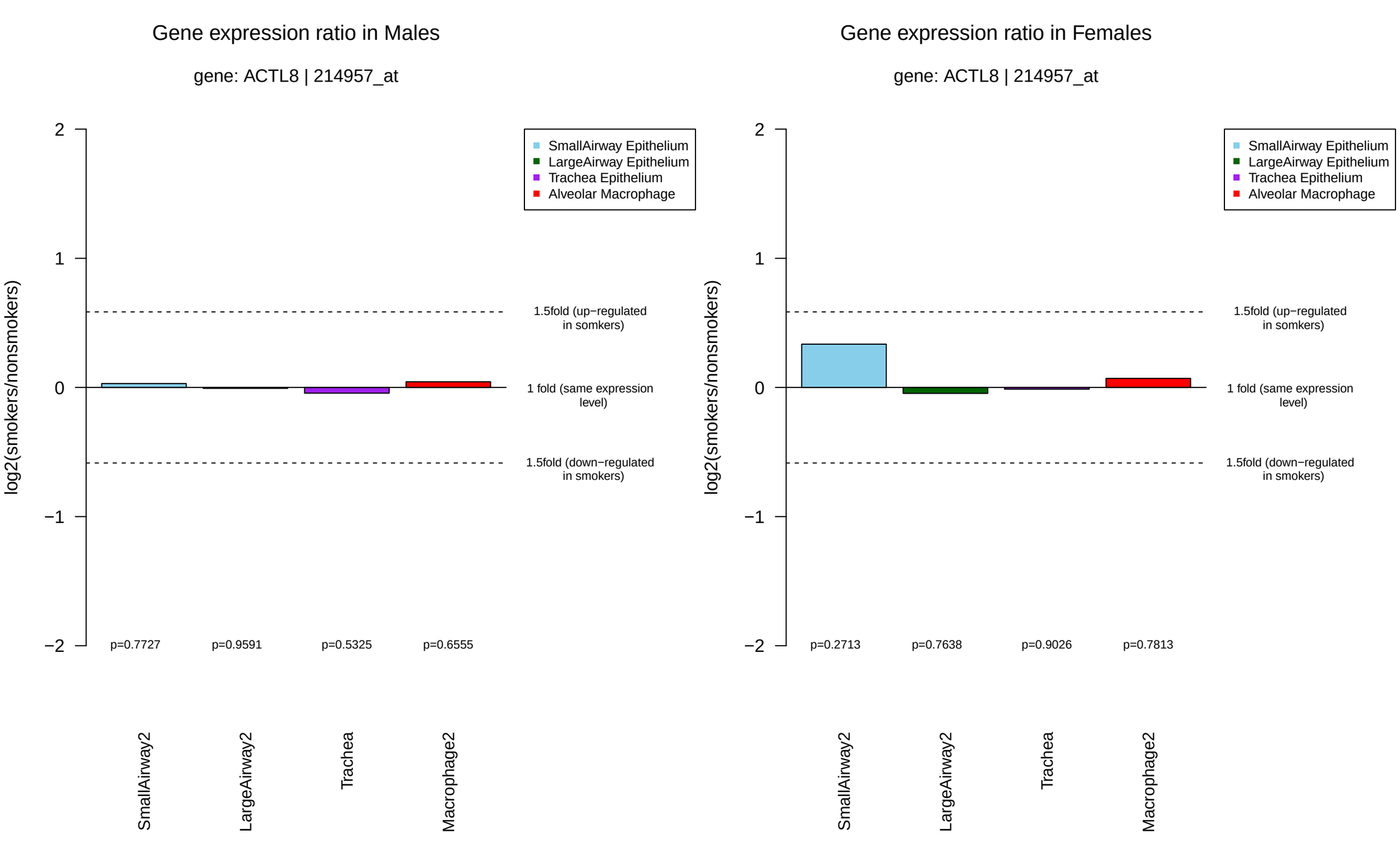
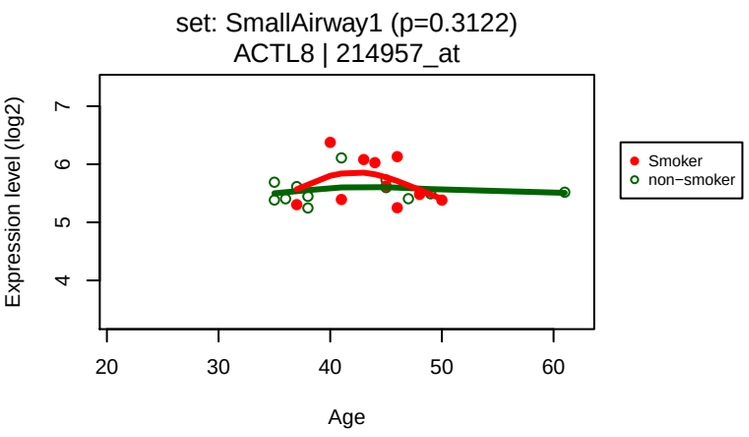
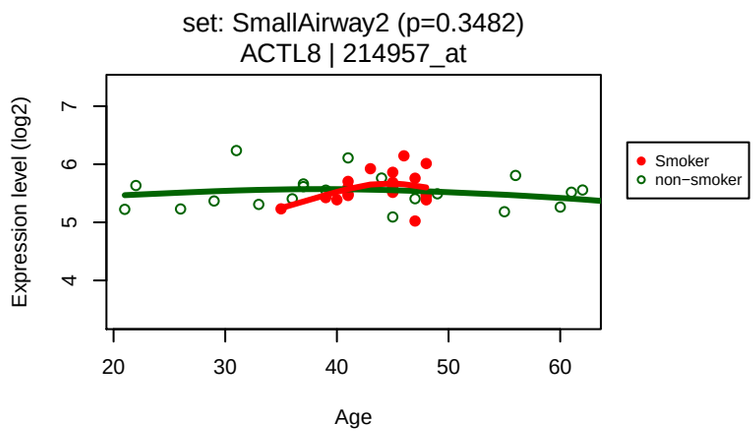
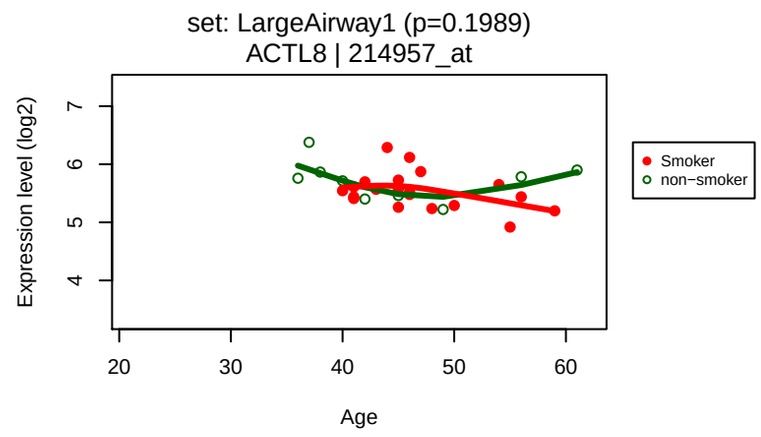

Gene expression of smokers and nonsmokers
gene: ACTL8 | 214957_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3122
p=0.3482
p=0.3459
p=0.1989
p=0.9879
p=0.6328
p=0.1227
p=0.6329
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

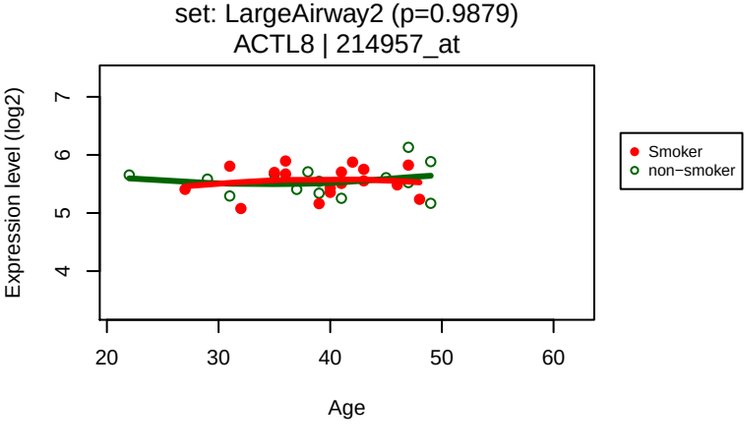

Gene expression ratio (smokers/nonsmokers)
gene: ACTL8 | 214957_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3122
p=0.3482
p=0.3459
p=0.1989
p=0.9879
p=0.6328
p=0.1227
p=0.6329
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

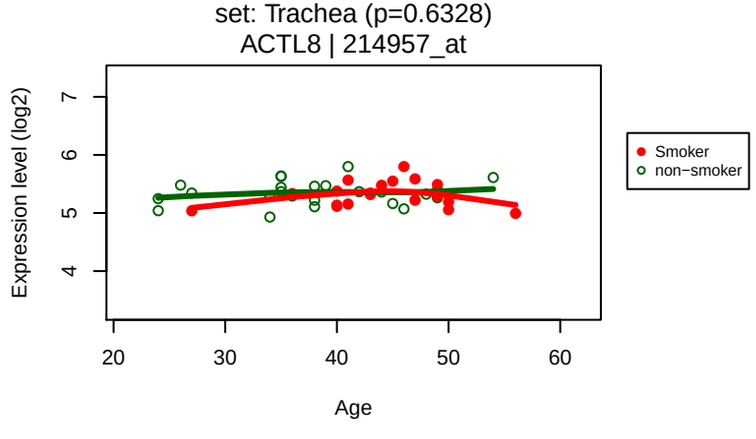

Gene expression ratio in Males
gene: ACTL8 | 214957_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7727
p=0.9591
p=0.5325
p=0.6555
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTL8 | 214957_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2713
p=0.7638
p=0.9026
p=0.7813
SmallAirway2
LargeAirway2
Trachea
Macrophage2

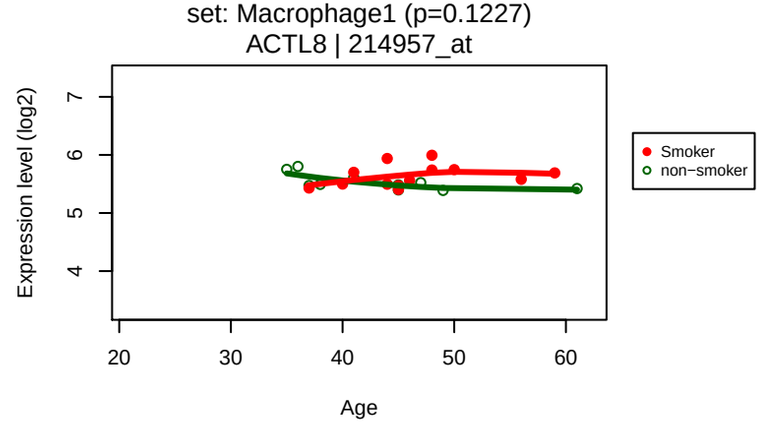

set: SmallAirway1 (p=0.3122)
ACTL8 | 214957_at
Expression level (log2)
Age
Smoker
non−smoker

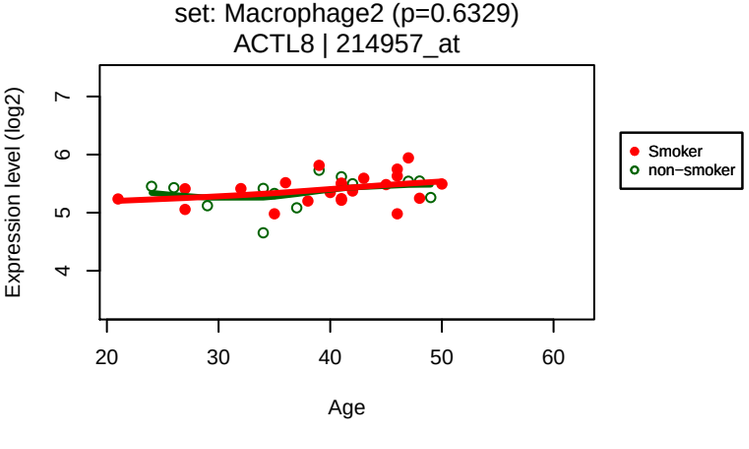

set: SmallAirway2 (p=0.3482)
ACTL8 | 214957_at
Expression level (log2)
Age
Smoker
non−smoker

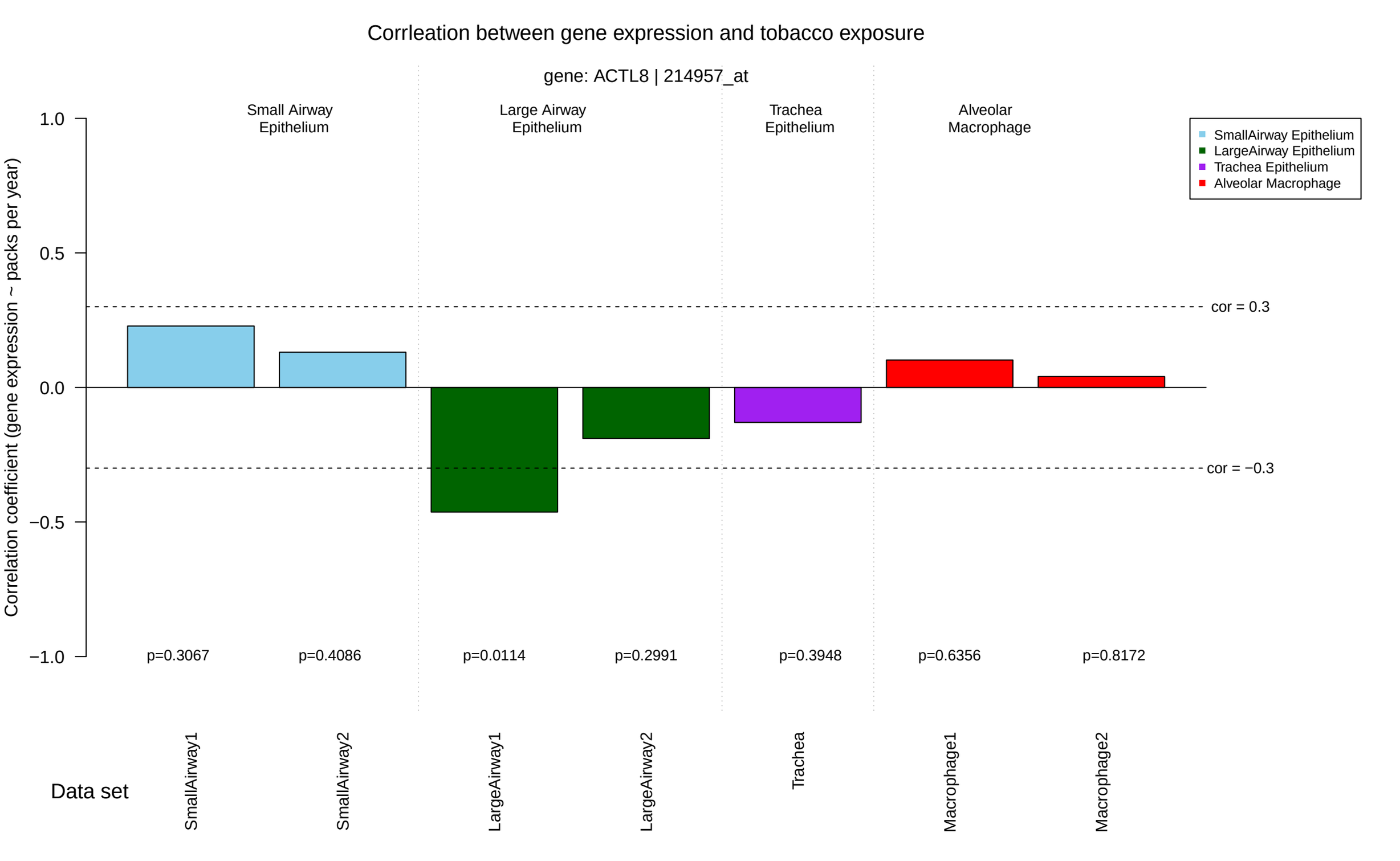

set: LargeAirway1 (p=0.1989)
ACTL8 | 214957_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9879)
ACTL8 | 214957_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6328)
ACTL8 | 214957_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1227)
ACTL8 | 214957_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6329)
ACTL8 | 214957_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTL8 | 214957_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3067
p=0.4086
p=0.0114
p=0.2991
p=0.3948
p=0.6356
p=0.8172
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2